16.8 Applying verifiable credentials across ACME’s products

The examples in earlier sections focused on an external auditor, but the same patterns naturally extend to ACME’s other authorization needs. In each case, verifiable credentials provide trusted context that ACME’s policy engine can evaluate without requiring ACME to create or manage accounts for every external participant. The PEP requests the attributes needed for policy evaluation, the holder responds with a presentation assembled from one or more credentials, and the PDP makes a decision using those normalized claims. The underlying issuance or presentation protocol may differ across ACME’s ecosystem, but the authorization flow stays consistent.

16.8.1 Supplier portal: Representing supplier authority across multi-tier relationships

ACME’s supplier portal enables manufacturers to manage contracts, purchase orders, invoices, and shipment records across their supply chains. These supply chains often involve multiple tiers: ACME provides a service platform to manufacturers, manufacturers manage on suppliers, and suppliers employ workers who need secure access to shared records. In the past, granting this access usually meant manufacturers had to create and manage accounts for each supplier’s workers, even though those workers weren't part of the manufacturer’s organization. This led to the familiar security problems of lingering accounts, inconsistent deprovisioning, and fragile onboarding processes as workers cycled through projects or changed employers. Verifiable credentials made this much simpler.

When a supplier employee signs in, ACME’s PEP sends an attribute query requesting claims to confirm the employee's employment at the supplier organization and their delegated authority for a particular contract or customer account. Most suppliers issue these credentials using OpenID4VCI, and employees present them through OpenID4VP flows.

The PEP verifies the credentials, confirms that the supplier is approved by the manufacturer, and forwards normalized attributes to the PDP. The